

MANISH CHOUDHARY

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PROFESSIONAL SUMMARY

Recent Computer Science & Engineering graduate (GNDEC 2026) with a 6-month Data Science internship building a dual-stage GenAI evaluation platform. Hands-on experience across the ML lifecycle -- data engineering, model training, API deployment, and explainability. Projects in medical imaging (0.983 AUROC), time-series forecasting (1.22% MAPE), and LLM-powered mentorship systems.

EDUCATION

Guru Nanak Dev Engineering College, Ludhiana, Punjab

2022 - 2026

B.Tech, Computer Science & Engineering

EXPERIENCE

Sabudh Foundation - Mohali, Punjab

Jul 2025 - Jan 2026

Data Science Intern

- Led a 5-member team to build a dual-stage GenAI platform, replacing 3 legacy assessment tools (Google Forms + KoboToolbox).
- Built an asynchronous LLM pipeline (Gemini API) that ingests multi-domain scores, synthesizes narrative insights, and serializes results to MySQL -- with a "Check-or-Create" cache achieving <0.5s retrieval for returning users.
- Designed a 7-table MySQL schema (students, 3 domain score tables, generated reports, API logs, assessment cycles) to replace a flat spreadsheet, reducing LLM token usage by isolating relevant domain data per query.
- Integrated Serper API for indirect RAG -- dynamically fetching verified course links based on AI-identified skill gaps, eliminating hallucinated URLs.
- Shipped a Streamlit app with NLP chatbot (multilingual: 22 Indian languages), admin dashboard, and automated PDF report cards (ReportLab) for 100+ NGO students.
- Delivered the NGO's first data-driven learning paths, replacing manual scoring and saving a weekly volunteer session.

PROJECTS

TB-MTNet: TB Detection and Severity Assessment

GitHub

Multimodal medical AI (Inception-v3 + ECA attention + Transformer encoder). Achieved **0.983 AUROC** for TB classification and **8.51 MAE** for severity regression across ~12K X-rays. Deployed with FastAPI, U-Net segmentation, and Grad-CAM explainability.

PyTorch - Inception-v3 - Transformers - FastAPI - U-Net - Grad-CAM

AI-Powered Stock Price Predictor

GitHub

4-layer LSTM model achieving **1.22% MAPE** (MSFT) and **1.92% MAPE** (AAPL). FastAPI inference service with Streamlit real-time dashboard.

TensorFlow - LSTM - FastAPI - Streamlit - Pandas

Integrated Learning Recommendation Engine

Dual-stage system: Stage I (Gemini API for semantic synthesis of competency scores) + Stage II (context-aware chatbot retrieving pre-computed insights from MySQL). Serper API web search for dynamic course curation, RBAC with password hashing, ReportLab PDF report cards.

Gemini API - Serper API - MySQL - Streamlit - ReportLab

CERTIFICATIONS

Sabudh Foundation -- Data Science Training

GitHub -- GitHub Foundations

RELEVANT COURSEWORK

Data Structures & Algorithms - Database Management Systems - Operating Systems - Computer Networks - Object-Oriented Programming - Machine Learning

SKILLS

Languages: Python, C/C++, SQL **ML / DL:** Scikit-Learn, TensorFlow, Keras, PyTorch, LSTM, Transfer Learning

GenAI: OpenAI API, Gemini API, Prompt Engineering, Hugging Face **Computer Vision:** OpenCV, YOLO, CNNs, Grad-CAM, U-Net

Data: Pandas, NumPy, MySQL, MongoDB **Web / API:** FastAPI, Streamlit **Tools:** Git, Docker, Label Studio